5 BLOOD AND CIRCULATION

Large, multicellular animals need a circulatory system to transport substances to and from the cells of the body. This chapter looks at the structure and function of the circulatory systems of humans and other animals, the composition of mammalian blood, and disorders associated with the heart and circulation.

LEARNING OBJECTIVES

- Understand why simple unicellular organisms can rely on diffusion for movement of substances in and out of the cell
- Understand the need for a transport system in multicellular organisms
- Understand the general structure of the circulation system, including the blood vessels to and from the heart and lungs, liver and kidneys
- Describe the structure of the heart and how it functions
- Understand how factors may increase the risk of developing coronary heart disease
- Explain how the heart rate changes during exercise and under the influence of adrenaline
- Understand how the structures of arteries, veins and capillaries relate to their functions
- Describe the composition of blood: red blood cells, white blood cells, platelets and plasma
- Understand the role of plasma in the transport of carbon dioxide, digested food, urea, hormones and heat energy
- Understand how the adaptations of red blood cells make them suitable for the transport of oxygen, including shape, the absence of a nucleus and the presence of haemoglobin
- Understand how the immune system responds to infection using white blood cells, illustrated by phagocytes ingesting pathogens and lymphocytes releasing antibodies specific to the pathogen

BIOLOGY ONLY

- Understand how vaccination results in the manufacture of memory cells, which enable future antibody production to the pathogen to occur sooner, faster and in greater quantity
- Understand how platelets are involved in blood clotting, which prevents blood loss and the entry of microorganisms

THE NEED FOR CIRCULATORY SYSTEMS

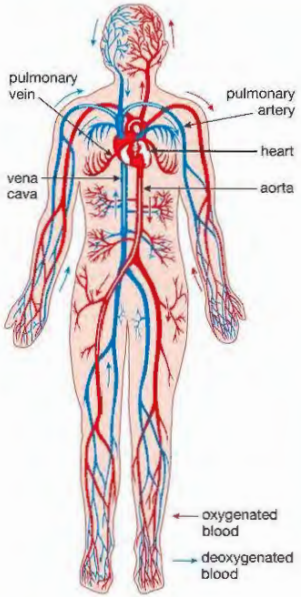
Figure 5.1 shows the circulatory system of a mammal.

Blood is pumped around a closed circuit made up of the heart and blood vessels. As it travels around the body, it collects materials from some places and unloads them in others. In mammals, blood transports:

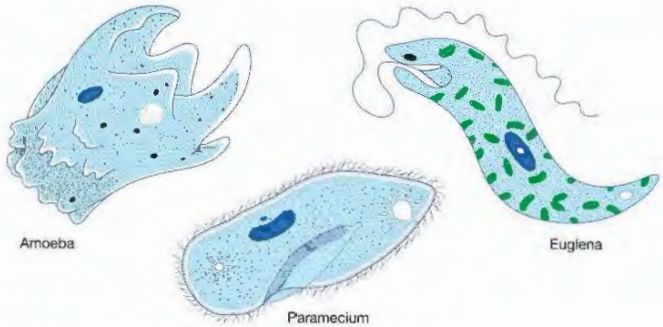
- oxygen from the lungs to all other parts of the body
- carbon dioxide from all parts of the body to the lungs
- nutrients from the gut to all parts of the body
- urea from the liver to the kidneys.

Hormones, antibodies and many other substances are also transported by the blood. It also distributes heat around the body.

Single-celled organisms, like the ones shown in Figure 5.2, do not have circulatory systems.



▲ Figure 5.1 The human circulatory system.



▲ Figure 5.2 Unicellular organisms do not have circulatory systems.

There is no circulatory system to carry materials around the very small 'bodies' of these single-celled organisms. Materials can easily move around the cell without a special system. There is no need for lungs or gills to obtain oxygen from the environment either. Single-celled organisms obtain oxygen by diffusion through the surface membrane of the cell. The rest of the cell then uses the oxygen. The area of the cell's surface determines how much oxygen the organism can get (the supply rate), and the volume of the cell determines how much oxygen the organism uses (the demand rate).

The ratio of supply to demand can be written as: $\frac{\text{surface area}}{\text{volume}}$

This is called the 'surface area to volume ratio' and it is affected by the size of an organism (see Chapter 1, Activity 5). Single-celled organisms have a high surface area to volume ratio. Their cell surface membrane has a large enough area to supply all the oxygen that their volume demands. In larger animals, the surface area to volume ratio is lower.

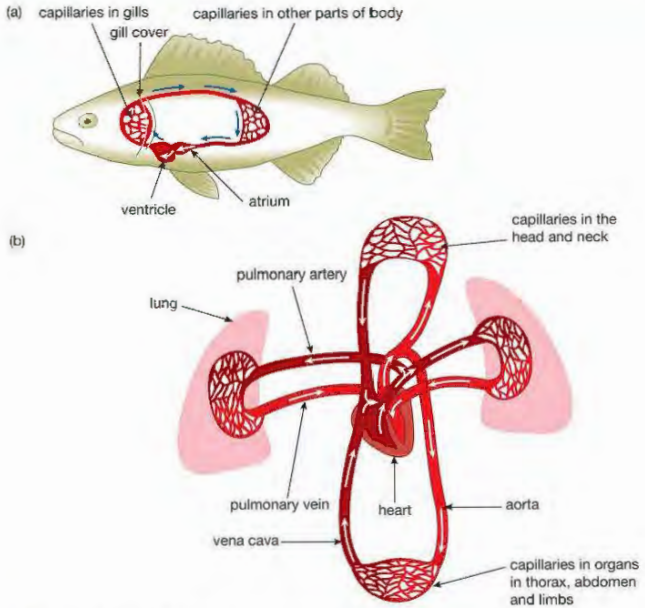
Large animals cannot get all the oxygen they need through their surface (even if the body surface would allow it to pass through) – there just isn't enough surface to supply all that volume. To overcome this problem, large organisms have evolved special gas exchange organs and circulatory systems. The gills of fish and the lungs of mammals are linked to a circulatory system that carries oxygen to all parts of the body. The same idea applies to obtaining nutrients – the gut obtains nutrients from food and the circulatory system distributes the nutrients around the body.

THE CIRCULATORY SYSTEMS OF DIFFERENT ANIMALS

One of the main functions of a circulatory system in animals is to transport oxygen. Blood is pumped to a gas exchange organ to load oxygen. It is then pumped to other parts of the body where it unloads the oxygen. There are two main types of circulatory systems in animals.

- In a **single circulatory** system the blood is pumped from the heart to the gas exchange organ and then directly to the rest of the body.
- In a **double circulatory** system the blood is pumped from the heart to the gas exchange organ, back to the heart and then to the rest of the body.

Figure 5.3 shows the difference between these systems.



▲ Figure 5.3 (a) The single circulatory system of a fish. The blood passes through the heart only once in a complete circuit of the body. (b) The double circulatory system of a human (and other mammals). The blood passes through the heart twice in one complete circuit of the body.

KEY POINT

'Pulmonary' means 'relating to the lungs'.

DID YOU KNOW?

There are actually two vena cavae (the plural of vena cava). One brings blood back from the head and arms and the other returns blood from the rest of the body (see Figure 5.4).

There are two parts to a double circulatory system:

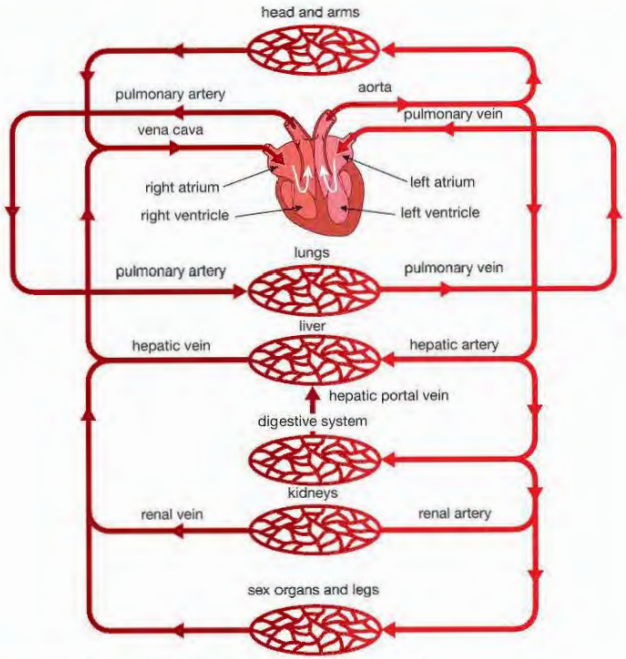
- The **pulmonary circulation**. Deoxygenated blood leaves the heart through the pulmonary arteries, and is circulated through the lungs, where it becomes oxygenated. The oxygenated blood returns to the heart through the pulmonary veins.
- The **systemic circulation**. Oxygenated blood leaves the heart through the aorta and is circulated through all other parts of the body, where it unloads its oxygen. Deoxygenated blood returns to the heart through the vena cava.

A double circulatory system is more efficient than a single circulatory system. The heart pumps the blood twice, so higher pressures can be maintained. The blood travels more quickly to organs. In the single circulatory system of a fish, blood loses pressure as it passes through the gills. It then travels more slowly to the other organs.

The human circulatory system comprises:

- the heart – this is a pump
- blood vessels – these carry the blood around the body; **arteries** carry blood away from the heart and towards other organs, **veins** carry blood towards the heart and away from other organs and **capillaries** carry blood through organs, linking the arteries and veins
- blood – the transport medium.

Figure 5.4 shows the main blood vessels in the human circulatory system.



▲ Figure 5.4 The main components of the human circulatory system

THE STRUCTURE AND FUNCTION OF THE HUMAN HEART

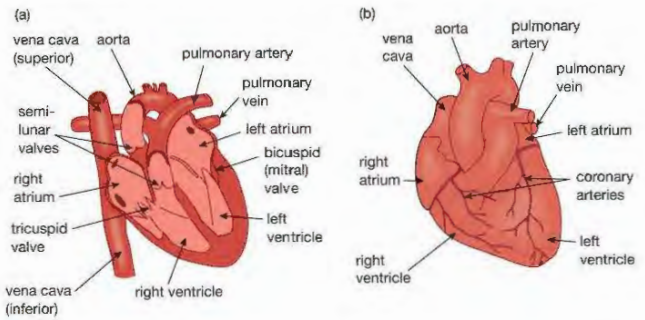
The human heart is a pump (Figure 5.5). It pumps blood around the body at different speeds and at different pressures according to the body's needs.

DID YOU KNOW?

'Cardiac' means 'related to the heart'.

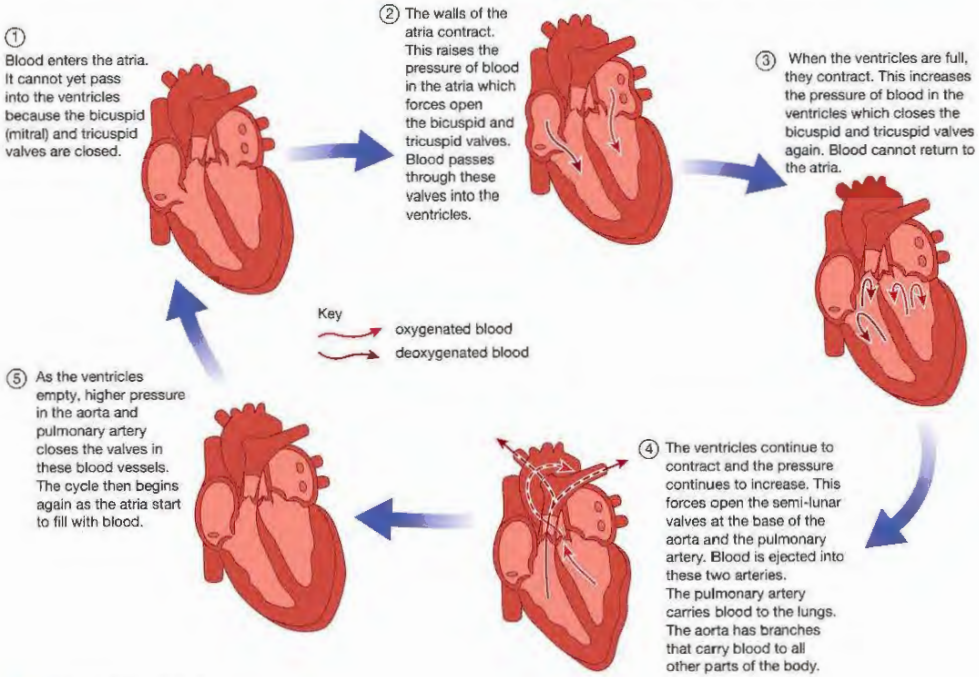
DID YOU KNOW?

The **bicuspid** (mitral) and **tricuspid valves** are both sometimes called atrioventricular valves, as each controls the passage of blood from an atrium to a ventricle.



▲ Figure 5.5 The human heart: (a) vertical section; (b) external view

Blood is moved through the heart by a series of contractions and relaxations of the muscle in the walls of the four chambers. These events form the **cardiac cycle**. The main stages are illustrated in Figure 5.6.



▲ Figure 5.6 The cardiac cycle.

HINT

Note that during the cardiac cycle both atria contract at the same time (and then relax). After this both ventricles contract at the same time (and then relax). Students are sometimes confused about this, and think that one ventricle contracts, followed by the other.

KEY POINT

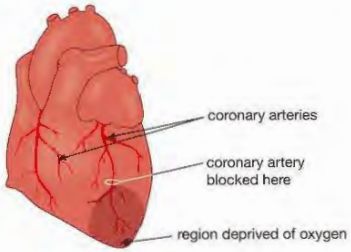
Cardiac muscle is unlike any other muscle in our bodies. It never gets fatigued ('tired') like skeletal muscle. On average, cardiac muscle fibres contract and then relax again about 70 times a minute. In a lifetime of 70 years, this special muscle will contract over two billion times – without taking a rest!

The structure of the heart is adapted to its function in several ways:

- It is divided into a left side and a right side by a wall of muscle called the septum. The right **ventricle** pumps blood only to the lungs while the left **ventricle** pumps blood to all other parts of the body. This requires much more pressure, which is why the wall of the left ventricle is much thicker than that of the right ventricle.
- Valves ensure that blood can flow only in one direction through the heart.
- The walls of the **atria** are thin. They can be stretched to receive blood as it returns to the heart but can contract with enough force to push blood through the bicuspid and tricuspid valves into the ventricles.
- The walls of the heart are made of **cardiac muscle**, which can contract and then relax continuously without becoming fatigued.
- The cardiac muscle has its own blood supply – the coronary circulation. Blood reaches the muscle via **coronary arteries**. These carry blood to capillaries that supply the heart muscle with oxygen and nutrients. Blood is returned to the right atrium via **coronary veins**.

CORONARY HEART DISEASE

The coronary arteries are among the narrowest in the body. They are easily blocked by a build-up of fatty substances (including **cholesterol**) in their walls. This can cut off the blood supply to an area of cardiac muscle. The affected muscle can no longer receive oxygen and glucose, so it cannot respire and release energy. This means it is unable to contract, resulting in a heart attack. This is called **coronary heart disease** (CHD). It can lead to severe health problems and is often fatal.



▲ Figure 5.7 A blockage of a coronary artery cuts off the blood supply to part of the heart muscle.

A number of factors make coronary heart disease more likely:

- heredity – some people inherit a tendency to develop coronary heart disease
- high blood pressure – puts more strain on the heart
- diet – eating large amounts of saturated fat is likely to raise cholesterol levels
- smoking – raises blood pressure and makes blood clots more likely to form
- stress – raises blood pressure
- lack of exercise – regular exercise helps to reduce blood pressure and strengthens the heart.

DID YOU KNOW?

Have you noticed a 'hollow' or 'fluttering' feeling in your stomach when you are anxious? It happens because blood that would normally flow to your stomach and intestines has been diverted to the muscles to allow the 'fight or flight' response.

HEART RATE

Normally the heart beats about 70 times a minute, but this can change according to the needs of the body. When we exercise, muscles must release more energy. They need an increased supply of oxygen for aerobic respiration (see Chapter 1). To deliver the extra oxygen, both the number of beats per minute (heart rate) and the volume of blood pumped with each beat (called stroke volume) increase.

When we are angry or afraid our heart rate again increases. The increased output supplies extra blood to the muscles, enabling them to release extra energy through aerobic respiration. This allows us to fight or run away and is called the 'fight or flight' response. It is triggered by **secretion** of the hormone **adrenaline** from the adrenal glands (see Chapter 7).

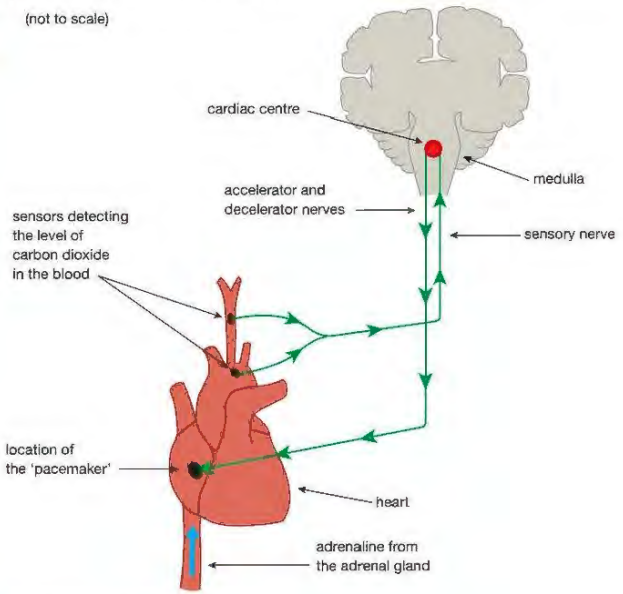
When we sleep, our heart rate decreases as all our organs are working more slowly. They need to release less energy and so need less oxygen.

These changes in the heart rate are controlled by nerve impulses from a part of the brain called the **medulla** (Figure 5.8). When we start to

exercise, our muscles produce more carbon dioxide in aerobic respiration. Receptors in the aorta and the carotid artery (the artery leading to the head) detect this increase. They send electrical signals called nerve impulses through the sensory nerve to the medulla. The medulla responds by sending nerve impulses along the accelerator nerve. When carbon dioxide production returns to normal, the medulla receives fewer impulses. It responds by sending nerve impulses along a decelerator nerve.

KEY POINT

The precise region of the medulla that controls heart functions is called the **cardiac centre**.



▲ Figure 5.8 How the heart rate is controlled.

The accelerator nerve increases the heart rate. It also causes the heart to beat with more force and so increases blood pressure. The decelerator nerve decreases the heart rate. It also reduces the force of the contractions. Blood pressure then returns to normal.

These controls are both examples of **reflex actions** (see Chapter 6).

ARTERIES, VEINS AND CAPILLARIES

Arteries carry blood from the heart to the organs of the body. This arterial blood is pumped out by the ventricles at a high pressure. Elastic tissue in the walls of the arteries allows them to stretch and recoil (spring back into shape), maintaining the high blood pressure. A thick muscular wall helps control the flow of blood by dilating (widening) or constricting (narrowing) the vessels.

Veins carry blood from organs back towards the heart. The pressure of this venous blood is much lower than that in the arteries. It puts very little pressure on the walls of the veins, so they can be thinner than those of arteries, and contain less elastic tissue and muscle. Figure 5.9 shows the structure of a typical artery and a typical vein with the same diameter.

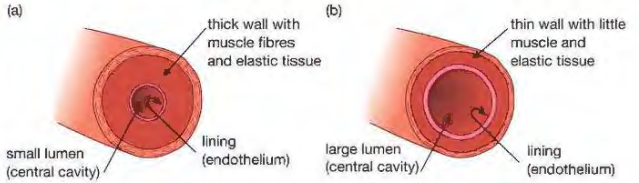
EXTENSION WORK

Arterioles are small arteries. They carry blood into organs from arteries. Their structure is similar to the larger arteries, but they have a larger proportion of muscle fibres in their walls. They are also supplied with nerve endings in their walls and so can be made to dilate (become wider) or constrict (become narrower) to allow more or less blood into the organ.

If *all* the arterioles constrict, it is harder for blood to pass through them – there is more resistance. This increases blood pressure. Prolonged stress can cause arterioles to constrict and so increase blood pressure.

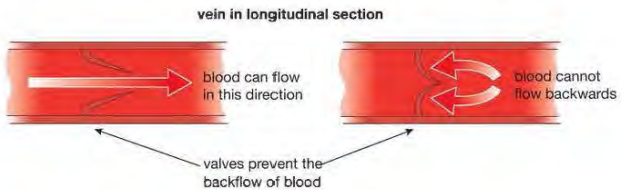
KEY POINT

All arteries carry oxygenated blood (blood containing a lot of oxygen) except the pulmonary artery and the umbilical artery of an unborn baby. All veins carry deoxygenated blood (blood containing less oxygen) except the pulmonary vein and umbilical vein.



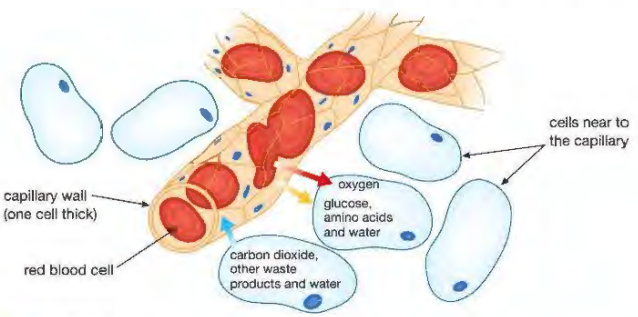
▲ Figure 5.9 The structure of (a) an artery and (b) a vein as seen in cross section.

Veins also have **semilunar** (half-moon shaped) **valves**, which prevent the backflow of blood. The action of these valves is explained in Figure 5.10.



▲ Figure 5.10 The action of semilunar valves in veins

Capillaries carry blood through organs, bringing the blood close to every cell in the organ. Substances are transferred between the blood in the capillary and the cells. To do this, capillaries must be small enough to 'fit' between cells, and allow materials to pass through their walls easily. Figure 5.11 shows the structure of a capillary and how exchange of substances takes place between the capillary and nearby cells. The walls of capillaries are one cell thick, providing a short distance for diffusion of materials into and out of the blood. Red blood cells just fit through the tiny diameter of capillaries, so they are close to the capillary wall. This means that there is a short distance for oxygen to diffuse. Figure 5.12 shows a photograph of a cross-section through an artery and a vein.



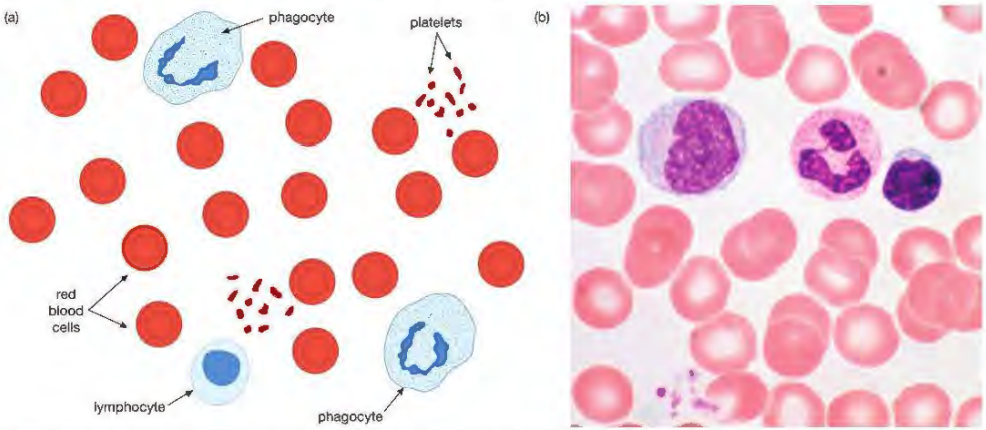
▲ Figure 5.11 How capillaries exchange materials with cells



▲ Figure 5.12 The lumen of the artery is the same size as the lumen of the vein – but note the difference in the thickness of the walls of these two vessels.

THE COMPOSITION OF BLOOD

Blood is a lot more than just a red liquid flowing through your arteries and veins! In fact, blood is a complex tissue. Figure 5.13 illustrates the main types of cells found in blood.



▲ Figure 5.13 The different types of blood cells. (a) Diagram of the different cells. (b) A blood smear seen through a light microscope. The smear shows many red blood cells and three different kinds of white blood cell.

The different parts of blood have different functions. These are described in Table 5.1.

Table 5.1 Functions of the different components of blood.

COMPONENT OF BLOOD	DESCRIPTION OF COMPONENT	FUNCTION OF COMPONENT
plasma	liquid part of blood: mainly water	carries the blood cells around the body; carries dissolved nutrients, hormones, carbon dioxide and urea; also distributes heat around the body
red blood cells (erythrocytes)	biconcave, disc-like cells with no nucleus; millions in each mm^3 of blood	transport of oxygen – contain mainly haemoglobin, which loads oxygen in the lungs and unloads it in other regions of the body
WHITE BLOOD CELLS:		
lymphocytes	about the same size as red cells with a large spherical nucleus	produce antibodies to destroy microorganisms – some lymphocytes persist in our blood after infection and give us immunity to specific diseases
phagocytes	much larger than red cells, with a large spherical or lobed nucleus	digest and destroy bacteria and other microorganisms that have infected our bodies
platelets	the smallest cells – are really fragments of other cells	release chemicals to make blood clot when we cut ourselves

RED BLOOD CELLS

The red blood cells or **erythrocytes** are highly specialised cells made in the bone marrow. They have a limited life span of about 100 days after which time they are destroyed in the spleen. They have only one function – to transport oxygen. Several features enable them to carry out this function very efficiently.

Red blood cells contain **haemoglobin**. This is an iron-containing protein that associates (combines) with oxygen to form **oxyhaemoglobin** when

there is a high concentration of oxygen in the surroundings. We say that the red blood cell is loading oxygen. When the concentration of oxygen is low, oxyhaemoglobin turns back into haemoglobin and the red blood cell unloads its oxygen.

As red blood cells pass through the lungs, they load oxygen. As they pass through active tissues they unload oxygen.

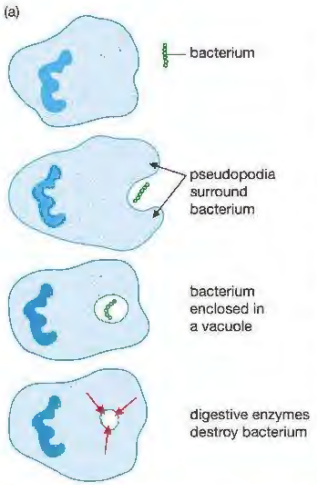
$$\text{haemoglobin} + \text{oxygen} \underset{\text{in the tissues}}{\overset{\text{in the lungs}}{\rightleftharpoons}} \text{oxyhaemoglobin}$$

Red blood cells do not contain a nucleus. It is lost during their development in the bone marrow. This means that more haemoglobin can be packed into each red blood cell so more oxygen can be transported. Their biconcave shape allows efficient exchange of oxygen in and out of the cell. Each red blood cell has a high surface area to volume ratio, giving a large area for diffusion. The thin shape of the cell results in a short diffusion distance to the centre of the cell.

WHITE BLOOD CELLS

There are several types of white blood cell. Their main role is to protect the body against invasion by disease-causing microorganisms (pathogens), such as bacteria and viruses. They do this in two main ways: **phagocytosis** and production of **antibodies**.

About 70% of white blood cells can ingest (take in) microorganisms such as bacteria. This is called phagocytosis, and the cells are **phagocytes**. They do this by changing their shape, producing extensions of their cytoplasm, called pseudopodia. The pseudopodia surround and enclose the microorganism in a vacuole. Once it is inside, the phagocyte secretes enzymes into the vacuole to break the microorganism down (Figure 5.14). Phagocytosis means 'cell eating' – you can see why it is called this.



▲ Figure 5.14 (a) Phagocytosis by a white blood cell. (b) A phagocyte engulfing a pathogenic microorganism (a species of yeast).

Approximately 25% of white blood cells are **lymphocytes**. Their function is to make chemicals called antibodies. Antibodies are soluble proteins that pass into the plasma. Pathogens such as bacteria and viruses have chemical 'markers' on their surfaces, which the antibodies recognise. These markers are called **antigens**. The antibodies stick to the surface antigens and destroy the pathogen. They do this in a number of ways, for example by:

- causing bacteria to stick together, so that phagocytes can ingest them more easily
- acting as a 'label' on the pathogen, so that it is more easily recognised by a phagocyte
- causing bacterial cells to burst open
- neutralising poisons (toxins) produced by pathogens.

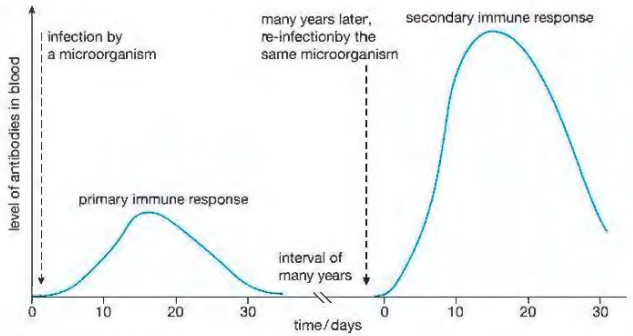
The production of antibodies following the first exposure to a foreign antigen is called the primary immune response.

BIOLOGY ONLY

IMMUNITY

Some lymphocytes do not get involved in killing microorganisms straight away. Instead, they develop into **memory cells**. These cells remain in the blood for many years, sometimes a lifetime. If the same microorganism re-infects a person, the memory lymphocytes start to reproduce and produce antibodies, so that the pathogen can be quickly dealt with. This is known as immunity.

This secondary **immune response** is much faster and more effective than the primary response. The number of antibodies in the blood quickly rises to a high level, killing the microorganisms before they have time to multiply to a point where they would cause disease. This is shown in Figure 5.15.

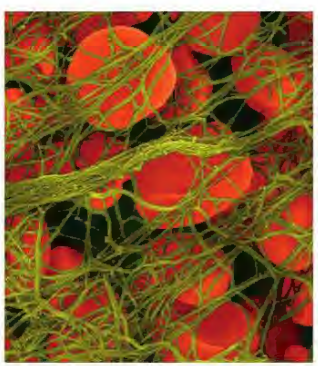


▲ Figure 5.15 The primary and secondary immune responses

A person can be given artificial immunity to a disease-causing organism without ever actually contracting (having) the disease itself. This is done by **vaccination**. A person is injected with an 'agent' that carries the same antigens as a specific pathogen. Lymphocytes recognise the antigens and multiply exactly as if that microorganism had entered the bloodstream. They produce memory cells and make the person immune to the disease. If the person now comes into contact with the 'real' pathogen, they will experience a secondary immune response. Antibody production will happen sooner, faster and in greater quantity than if they had not been vaccinated, and may be enough to prevent the pathogen reproducing in the body and causing the disease.

Some agents used as vaccines are:

- a weakened strain of the actual microorganism, e.g. vaccines against polio, tuberculosis (TB) and measles
- dead microorganisms, e.g. typhoid and whooping cough vaccines
- modified toxins of the bacteria, e.g. tetanus and diphtheria vaccines
- just the antigens themselves, e.g. influenza vaccine
- harmless bacteria, genetically engineered to carry the antigens of a different, disease-causing microorganism, e.g. the vaccine against hepatitis B.



▲ Figure 5.16 Red blood cells trapped in fibres of fibrin, forming a blood clot.

PLATELETS

Platelets are not whole cells, but fragments of large cells made in the bone marrow. If the skin is cut, exposure to the air stimulates the platelets and damaged tissue to produce a chemical. This chemical causes the soluble plasma protein **fibrinogen** to change into insoluble fibres of another protein, **fibrin**. The fibrin forms a network across the wound, in which red blood cells become trapped (Figure 5.16) This forms a clot, which prevents further loss of blood and entry of microorganisms that may be pathogens. The clot develops into a scab, which protects the damaged tissue while new skin grows.

END OF BIOLOGY ONLY

CHAPTER QUESTIONS

More questions on blood and circulation can be found at the end of Unit 2 on page 130

SKILLS CRITICAL THINKING

6 1 After a period of exercise, which blood vessel will contain the highest concentration of carbon dioxide?

A aorta
B vena cava
C hepatic artery
D pulmonary vein

2 When the right ventricle contracts, to which of the following structures does the blood flow next?

A aorta
B left atrium
C pulmonary artery
D left ventricle

3 The diagram below shows sections through three blood vessels (not drawn to scale).



Which row in the table shows the correct names of vessels X, Y and Z?

	X	Y	Z
A	vein	capillary	artery
B	artery	capillary	vein
C	vein	artery	capillary
D	capillary	vein	artery

5 4 Which component of the blood makes antibodies?

A red blood cells
B white blood cells
C plasma
D platelets

6 5 Some animals have a single circulatory system, some have a double circulatory system and some organisms have no circulatory system at all.

a Name one type of animal with a single circulatory system and one type of animal with a double circulatory system.

SKILLS REASONING

6 b Explain:

i the difference between single and double circulatory systems

7 ii why a double circulatory system is more efficient than a single circulatory system.

6 c Explain why single-celled organisms do not need a circulatory system.

SKILLS CRITICAL THINKING (6)

6 Blood transports oxygen and carbon dioxide around the body. Oxygen is transported by the red blood cells.

a Give three ways in which a red blood cell is adapted to its function of transporting oxygen.

b Describe how oxygen:

(5) i enters a red blood cell from the alveoli in the lungs

ii passes from a red blood cell to an actively respiring muscle cell.

(6) c Describe how carbon dioxide is transported around the body.

SKILLS REASONING (6)

7 Blood is carried around the body in arteries, veins and capillaries.

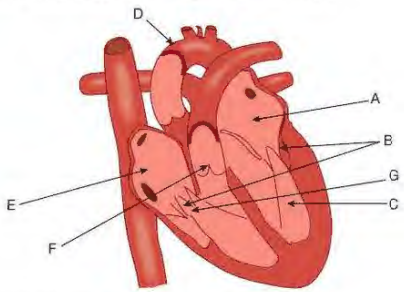
a Describe two ways in which the structure of an artery is adapted to its function.

b Describe three differences between arteries and veins.

c Describe two ways in which the structure of a capillary is adapted to its function.

SKILLS ANALYSIS

8 The diagram shows a section through a human heart.



(4) a Name the structures labelled A, B, C, D and E.

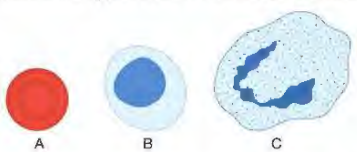
SKILLS CRITICAL THINKING (6) b What is the importance of the structures labelled B and F?

c Which letters represent the chambers of the heart to which blood returns:

SKILLS ANALYSIS (5) i from the lungs

ii from all the other organs of the body?

9 The diagram shows three types of cells found in human blood.



a Giving a reason for each answer, identify the blood cell which:

i transports oxygen around the body

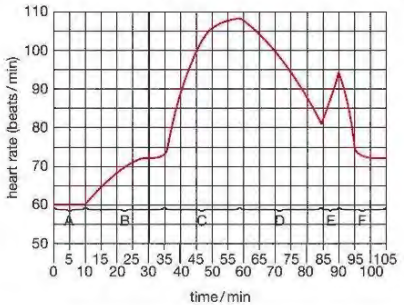
ii produces antibodies to destroy bacteria

iii engulfs and digests bacteria.

SKILLS CRITICAL THINKING b Name one other component of blood found in the plasma and state its function.

SKILLS ANALYSIS (6)

10 The graph shows changes in a person's heart rate over a period of time.



Giving reasons for your answers, give the letter of the time period when the person was:

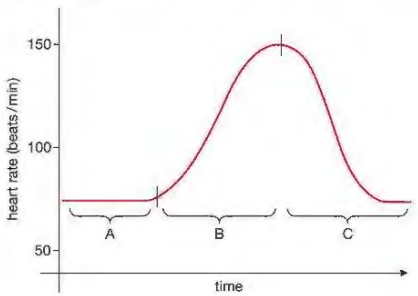
a running

b frightened by a sudden loud noise

c sleeping

d waking.

11 The graph shows the changes that take place in heart rate before, during and after a period of exercise.



a Describe and explain the heart rates found:

i at rest, before exercise (period A)

ii as the person commences the exercise (period B)

iii as the person recovers from the exercise (period C).

SKILLS REASONING (7) b How can the recovery period (period C) be used to assess a person's fitness?